Sahaj Bhadja

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Summary

Backend software engineer pursuing an MSCS at USC with 2+ years building **microservices**, **REST APIs**, and production distributed systems. Strong in **Go**, concurrency, databases, and **Docker**/cloud deployments with well-tested, maintainable code in fast-paced environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Computer Networks, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Structures

Experience

USC Dornsife **Los Angeles, CA**
Backend Software Engineer *September 2025 – Present*
- Designed and implemented backend features end to end for a multi-tenant platform using FastAPI, **PostgreSQL**, and Azure services, optimizing queries and indexing to reduce dashboard latency from **1.2s to 350ms** across distributed production workloads.
- Built fault-tolerant asynchronous job orchestration with explicit concurrency controls processing **1K+ jobs**, debugging failures across **distributed systems** and shipping well-tested CI/CD releases achieving **99%+ reliability** for 500+ users.

Grad8 **Los Angeles, CA**
Backend Software Engineer Intern *June 2025 – August 2025*
- Developed production **REST APIs** and Flask microservice modules on **AWS RDS** serving **10K+ daily requests**, implementing concurrent batch processing with multithreading and writing unit-tested code for instructor-facing grading workflows at scale.
- Contributed to API design and service communication patterns across dashboard, auth, and grading services, collaborating with product engineers to debug issues, resolve performance bottlenecks, and deliver maintainable backend enhancements under peak load.

Aasma Technology Solutions **India**
Backend Software Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform **microservices** with large-scale **REST API** integrations and **AWS Glue**/S3 pipelines, improving response times by **30%** while maintaining **20+ unit and integration test suites** in Jenkins for production-grade distributed data-sharing systems.
- Debugged service-to-service issues across Windows, Linux, and macOS deployments, contributing to system design discussions on API pagination, caching, and ETL efficiency that reduced technical debt by **20%** in Agile Scrum teams.

Aasma Technology Solutions **India**
Backend Software Engineer Intern - (VibrantChem) *May 2023 – August 2023*
- Implemented **8 microservices** for authentication, billing, and resource allocation with **REST APIs** and WebSockets syncing **2M+ records**, defining service boundaries and asynchronous communication patterns for scalable, maintainable backend architecture.
- Deployed **Docker** containerized services through Jenkins CI/CD on Azure App Services, debugging integration failures and optimizing database latency to cut deployment cycles by **50%** in a fast-paced production environment.

LST Capitals **India**
Backend Software Engineer Intern *November 2022 – January 2023*
- Built Flask **REST APIs** for booking and payment workflows across **AWS**-hosted microservices and Firebase backends, delivering sub-**200ms** latency on **50K+ records** with webhook-driven service-to-service communication and structured error handling.
- Debugged distributed payment and synchronization issues across mobile and web clients, writing clean maintainable backend modules with unit automation that streamlined cross-platform delivery by **22%** under tight deadlines.

Skills

Backend & Go: **Golang**, goroutines, asynchronous development, microservices, REST APIs, messaging systems (familiar), API design, service-to-service communication, concurrency, networking, TCP/IP, distributed systems, system design
Languages & Data: Python, Java, JavaScript, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Azure SQL, query optimization, performance tuning, object-oriented design
DevOps & Cloud: Docker, Kubernetes (familiar), AWS (EC2, S3, RDS, Glue, ECS), Azure, GCP (familiar), CI/CD, Jenkins, GitHub Actions, Git, Linux, unit testing, integration testing, problem solving, Agile

Projects

GoGateway Code | *Golang, Goroutines, REST APIs, Docker*
- Built production-style **Go microservices** with **goroutine**-based workers, **REST API** routing, and inter-service HTTP communication, containerized with **Docker** and covered by unit tests for auth, rate limiting, and request validation modules.

Linguistix | *AWS, Docker, ECS/Fargate, REST APIs, CI/CD*
- Deployed a containerized data-processing API on **AWS CodePipeline** and **ECS/Fargate** with automated CI/CD releases and validation checks, demonstrating scalable microservice deployment patterns and production-grade service reliability.

FinNexus Live | *TypeScript, WebSockets, MongoDB, Distributed Backend*
- Crafted a scalable fintech backend supporting **1M+ transactions** with real-time WebSocket updates, secure tokenized APIs, and advanced filtering, debugging performance issues across high-volume distributed data access patterns.